

A FAITHFUL KREĬN-SPACE REALIZATION OF THE WEIL QUADRATIC FORM, WITH A NO-GO THEOREM FOR THE ZERO-SIDE ATTACK ON POSITIVITY

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ABSTRACT. Weil’s criterion identifies the Riemann Hypothesis (RH) with the positive semidefiniteness of the Weil functional $\mathfrak{t}$ read from the Riemann–Weil explicit formula. We give a rigorous operator-theoretic realization of $\mathfrak{t}$ on the classical band-limited (Paley–Wiener) test class PW_d , $d > \frac{1}{2}$, in three layers. **(A) Semibounded realization.** Modulo a single, precisely isolated hypothesis (H)—a uniform short-interval zero-density statement strictly weaker than RH—the form $\mathfrak{t}$ is closable and defines a self-adjoint operator $\mathcal{T}$ that is bounded below, with $\inf \text{spec}(\mathcal{T})$ a finite, convergent quantity; consequently $\text{RH} \iff \inf \text{spec}(\mathcal{T}) \geq 0$. Four supporting lemmas are unconditional. **(B) Kreĭn structure.** We prove the exact identity $\mathfrak{t} = E^*JE$, where E is evaluation at the (near-line) zeros and J is a fundamental symmetry whose negative directions are exactly the off-line zeros; the construction is therefore a Kreĭn-space problem (indefinite metric), not a de Branges (a priori positive) one. The angular operator K of the sampling range yields a single-axis dictionary: $\|K\| < \infty \iff$ finiteness of the bottom, and $\|K\| \leq 1 \iff \text{RH}$. **(C) No-go theorem.** We prove that the explicit dependence of K on the off-line depths carries no leverage toward the sign: the zero-side expression and the arithmetic side of the explicit formula are the same functional, so bounding $\|K\| \leq 1$ from the zero side is identically the problem of proving Weil positivity. We situate the construction relative to the Connes–Consani archimedean trace approach and the Suzuki de Branges completion: ours is strongly analogous, the distinguishing feature being its unconditional, explicitly Kreĭn, finite sampling ladder. We do *not* claim progress past the sign barrier; we claim a precise localization of it. This paper closes the explicit-formula/zero-side line of this line of work and isolates the only remaining route to RH (structural positivity).

CONTENTS

1. Introduction	2
1.1. Weil’s criterion and the sign problem	2
1.2. The three layers and the main results	2
1.3. Relation to prior work and honest scope	2
2. Setup and notation	3
3. The four unconditional lemmas	3
4. Coercivity, the depth split, and the hypothesis (H)	4
4.1. The honest status of (H)	4
5. Theorem A: the semibounded realization	5
6. Theorem B (i): the Kreĭn identity	5
7. Theorem B (ii): the angular operator and the single axis	6
7.1. Where the negativity lives, and a warning	6
8. Theorem C: a no-go theorem for the zero-side attack	6
9. Relation to prior work	7
10. Honest status and open problems	8

References

1. INTRODUCTION

1.1. Weil's criterion and the sign problem. Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ be the completed Riemann zeta function, with nontrivial zeros ρ . For a suitable even test function the Riemann–Weil explicit formula equates a sum over zeros with an archimedean term, a pole term, and a sum over primes. Polarizing, one obtains a Hermitian quadratic functional $\mathfrak{t}(F)$ with the property

$$(1) \quad \boxed{\text{RH} \iff \mathfrak{t}(F) \geq 0 \text{ for all admissible } F.}$$

This is Weil's positivity criterion [12, 3]. In this formulation the difficulty of RH is *entirely the sign* of $\mathfrak{t}$: the right-hand side of the explicit formula is indefinite. On-line zeros contribute $|F(\gamma)|^2 \geq 0$, while off-line zeros (should they exist) contribute terms of *either* sign. Proving the sign is proving RH; no weaker statement about the *magnitude* of $\mathfrak{t}$ suffices. This paper makes that precise and operator-theoretic, and then proves a structural theorem explaining why the most natural (zero-side) realization cannot, by itself, decide the sign.

1.2. The three layers and the main results. We work on the band-limited class PW_d (Paley–Wiener, exponential type $d > \frac{1}{2}$, L^2 on the line). Band-limited functions are entire, so the off-axis evaluations $F(\gamma + ib)$ forced by off-line zeros are automatic and L^2 -controlled; this is the technically clean home for Weil's criterion.

Theorem 1.1 (Semibounded realization; §5). *Assume (H) (Definition 4.1). On $PW_d \cap H^1$, $d > \frac{1}{2}$, the Weil form $\mathfrak{t}$ is closable in H_+ and its self-adjoint operator $\mathcal{T}$ satisfies $\inf \text{spec}(\mathcal{T}) \geq 1 - 4C > -\infty$ for a finite $C = C(d, \delta, \eta)$. Consequently $\text{RH} \iff \inf \text{spec}(\mathcal{T}) \geq 0$.*

Theorem 1.2 (Kreĭn identity and the single axis; §§6–7). *Unconditionally, $\mathfrak{t} = E^*JE$ with $J = J^*$, $J^2 = I$. The angular operator K of the closed sampling range is defined (type-density uniqueness), and*

$$\|K\| < \infty \iff (\text{finite bottom, } B\text{-2}), \quad \|K\| \leq 1 \iff \text{RH},$$

with $\|K\| \geq e^{d/2} > 1$ from the magnitude side.

Theorem 1.3 (No-go; §8). *The off-line depths $\{b_\rho\}$ enter $\|K\|$ only through $\mathfrak{t}$ itself. Hence bounding $\|K\| \leq 1$ from the zero side is identically Weil positivity, and every zero-position-free, sign-agnostic bound gives $\|K\|^2 \gtrsim e^d$ (the wrong side of 1). The zero-side realization, being defined in terms of the unknown off-line zeros, cannot attack the sign.*

1.3. Relation to prior work and honest scope. The reduction “ $\|K\| \leq 1 \iff \text{RH}$ ” is strongly analogous to the device of Connes–Consani [4], who root Weil positivity in the compression of the scaling action onto Sonin's space and control a residual operator by an eigenvalue-below-1 argument. The completion of the Weil form into a reproducing-kernel space is the theme of Suzuki [11], who obtains a *de Branges* space under RH. Our construction differs in being *unconditional* (no RH assumed; the metric is genuinely indefinite, hence Kreĭn, not de Branges) and in exposing an explicit finite sampling ladder. We do *not* claim an identification of our K with the Connes–Consani operator; Theorem 1.3 in fact explains why the zero-side realization, unlike the archimedean one, cannot serve as a computational handle.

Theorem 1.1 is conditional on (H); (H) is much weaker than RH but is not established unconditionally by current methods (§4.1). Theorems 1.2 and 1.3 are unconditional. **None of the three proves RH**, and Theorem 1.3 proves that this particular route cannot. The value of the paper is a clean, faithful reformulation together with a rigorous map of the wall, and the identification of the only remaining route (structural positivity, §10).

2. SETUP AND NOTATION

Test class. For $d > \frac{1}{2}$ let $PW_d = \{F \in L^2(\mathbb{R}) : \widehat{F} \in L^2[-d, d]\}$ be the Paley–Wiener space (entire functions of exponential type $\leq d$ in L^2), with $\widehat{F}(\xi) = \int F(t)e^{-it\xi} dt$. We use the dense domain $\mathcal{D} := PW_d \cap H^1$ (so $\widehat{F} \in H^1[-d, d] \hookrightarrow C^0$). Two classical inequalities recur:

$$(2) \quad (\text{Bernstein}) \quad \|F'\|_2 \leq d \|F\|_2, \quad (\text{Nikolskii}) \quad \|F\|_\infty \leq C_d \|F\|_2.$$

Zeros, depths, partners. Write a nontrivial zero as $\rho = \frac{1}{2} + b_\rho + it_\rho$, $b_\rho, t_\rho \in \mathbb{R}$, so $b_\rho = 0 \iff \rho$ on the critical line. The functional equation gives the symmetric quartet $\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$. Define the reflected partner involution

$$(3) \quad \sigma(\rho) := 1 - \bar{\rho} = \frac{1}{2} - b_\rho + it_\rho, \quad \sigma^2 = \text{id}, \quad b_{\sigma(\rho)} = -b_\rho, \quad t_{\sigma(\rho)} = t_\rho.$$

The zero density is $\rho(T) = \frac{1}{2\pi} \log \frac{T}{2\pi}$, and $S(T) = N(T) - \bar{N}(T) = O(\log T)$ unconditionally. The Nyquist cell has length $1/d$ and contains $\sim \rho(T)/d \rightarrow \infty$ zeros (over-sampling).

Weil functional. For $F \in \mathcal{D}$ let $h(z) := F(z)\overline{F(\bar{z})}$. The explicit formula gives

$$(4) \quad \mathfrak{t}(F) := \sum_{\rho} h(\text{spectral point of } \rho) = \underbrace{\mathfrak{a}(F)}_{\text{arch.}} + \underbrace{\mathfrak{q}(F)}_{\text{pole}} - \underbrace{\mathfrak{p}(F)}_{\text{primes}},$$

the spectral point of $\rho = \frac{1}{2} + b_\rho + it_\rho$ being $t_\rho + ib_\rho$ (on-line zeros sampled on the real axis). Grouping by partners (3) gives the on/off decomposition

$$(5) \quad \mathfrak{t}(F) = \underbrace{\sum_{b_\rho=0} |F(t_\rho)|^2}_{\mathfrak{t}_+(F) \geq 0} + \underbrace{\sum_{\text{off-line pairs}} 2 \operatorname{Re}[F(t_\rho + ib_\rho)\overline{F(t_\rho - ib_\rho)}]}_{\mathfrak{t}_-(F) \text{ indefinite}}.$$

With $\tilde{\mathfrak{t}}_-(F) := \sum_{b_\rho \neq 0} |F(t_\rho + ib_\rho)|^2$ one has $|\mathfrak{t}_-(F)| \leq 2\tilde{\mathfrak{t}}_-(F)$ (sharpened to an exact projection identity in §7).

Realization space. Let H_+ be the completion of $\mathcal{D}$ in the seminorm $\|F\|_{H_+}^2 := \mathfrak{t}_+(F)$ (the on-line sampling space). $\mathcal{T}$ is the self-adjoint operator associated with the form $\mathfrak{t}$ on H_+ .

3. THE FOUR UNCONDITIONAL LEMMAS

Lemma 3.1 (Integration by parts). *For $F \in \mathcal{D}$ and $g := |F|^2$, $E_g := \sum_{b_\rho=0} g(t_\rho) - \int g \rho dt = -\int g'(t) S(t) dt$.*

Proof. $\widehat{F} \in H^1 \hookrightarrow C^0$, and IBP in ξ gives $|F(t)| = O(1/t)$, so $g = O(t^{-2})$ and $g \log t \rightarrow 0$; the boundary terms $[gS]_{\pm\infty}$ vanish, and $dS = dN - \rho dt$ has locally bounded variation, so Stieltjes IBP yields the claim. $\square$

Lemma 3.2 (The quadrature error is $O(\mathfrak{t}_+)$). $|E_g| \leq 2d(\sup_{\text{supp } g} |S|) \|F\|_2^2 = O(\rho \|F\|_2^2) = O(\mathfrak{t}_+(F))$.

Proof. $g' = 2 \operatorname{Re}(F'\overline{F})$, so $\int |g'| \leq 2\|F'\|_2 \|F\|_2 \leq 2d\|F\|_2^2$ by Bernstein (2). With $|S| = O(\log T_0)$ on $\text{supp } g$ and the tail controlled by $g' = O(t^{-3})$, $|E_g| \leq C_d(\log T_0) \|F\|_2^2$. As the on-line main term is $\sim \rho \|F\|_2^2 \sim \frac{\log T_0}{2\pi} \|F\|_2^2$, this is $O(\mathfrak{t}_+)$. $\square$

Remark 3.3. B-2 (a finite bottom) needs only $E_g = O(\mathfrak{t}_+)$; the sharp $o(\mathfrak{t}_+)$ —which would require Montgomery pair-correlation / RH-level input—is *not* needed. This is the single most important calibration of this line of work: we need an O , not an o .

Lemma 3.4 (Vertical L^2 bound). *For $F \in PW_d$ and $|b| < \frac{1}{2}$, $\|F(\cdot + ib)\|_2^2 = \int_{-d}^d |\widehat{F}(\xi)|^2 e^{-2b\xi} d\xi \leq e^{2d|b|} \|F\|_2^2 \leq e^d \|F\|_2^2$.*

Proof. Parseval on the shifted line, then $e^{-2b\xi} \leq e^{2d|b|} \leq e^d$ on $|\xi| \leq d$, $|b| < \frac{1}{2} < d$. Exact, no RH. $\square$

The amplification $\kappa(b) := e^{2d|b|}$ (> 1 for $b \neq 0$) is the quantitative source of the lower bound $\|K\| \geq e^{d/2}$ in §7.

Lemma 3.5 (Off-line sum bound; Plancherel–Pólya). $\tilde{\mathbf{t}}_-(F) = \sum_{b_\rho \neq 0} |F(t_\rho + ib_\rho)|^2 \leq C_d \rho \|F\|_2^2$.

Proof. Off-line spectral points lie in the strip interior $|b_\rho| < \frac{1}{2} < d$ (margin $d - \frac{1}{2} > 0$). The band-limited Plancherel–Pólya/Bessel inequality, valid for arbitrary (possibly clustered) sequences in a strip interior, gives $\sum_n |F(w_n)|^2 \leq B \|F\|_2^2$ with $B \asymp_d \sup_x \#\{n : \operatorname{Re} w_n \in [x, x + \frac{1}{d}]\}$. The pointwise bound $|F(w)|^2 \leq C_d \int_{|x - \operatorname{Re} w| < 1/d} |F|^2$ summed over the cell count gives $B = O(\rho)$ by Riemann–von Mangoldt and $S = O(\log)$. $\square$

Remark 3.6 (Why $O(\rho)$ and not $o(\rho)$). Zero-density makes off-line zeros globally sparse, but they may cluster locally; the cell count is bounded by the total $O(\rho)$, not by the off-line density. Replacing it by the latter would give $o(\mathbf{t}_+) = \text{RH}$ —forbidden by possible clustering.

4. COERCIVITY, THE DEPTH SPLIT, AND THE HYPOTHESIS (H)

By Lemmas 3.4–3.5, $\tilde{\mathbf{t}}_- \leq C_d \rho \|F\|_2^2$. To conclude $\tilde{\mathbf{t}}_- \leq C \mathbf{t}_+$ (which is B-2) we need *coercivity* $\mathbf{t}_+(F) \geq c \rho \|F\|_2^2$, a lower sampling bound. This is not free: by the Beurling gap theorem the on-line zeros are a set of sampling for PW_d only if they have no gap exceeding π/d ; the density condition $D^- \geq d/\pi$ is necessary but not sufficient. A single Nyquist gap containing a bump of F destroys coercivity.

Depth split. Coercivity is needed only where $\tilde{\mathbf{t}}_- > 0$, i.e. near off-line zeros. Split off-line zeros by depth at a threshold δ : *shallow* ($|b_\rho| < \delta$, $\kappa \leq e^{2d\delta} = 1 + O(\delta)$, sample like on-line points) vs *deep* ($|b_\rho| \geq \delta$, amplified, the only danger). Thus B-2 reduces to: deep off-line zeros do not dominate any Nyquist cell at large height.

Hypothesis 4.1 (Uniform local sparsity of deep off-line zeros). There exist $\delta > 0$, $\eta > 0$, T_0 such that for all $T \geq T_0$ and every interval I of length $1/d$ centred at height T ,

$$\#\{\rho : |b_\rho| < \delta, t_\rho \in I\} \geq \eta \rho(T) |I|.$$

Equivalently, deep off-line zeros occupy at most $(1 - \eta)$ of each Nyquist cell.

Lemma 4.2 ((H) $\Rightarrow$ coercivity). *Under (H), every Nyquist cell contains $\geq \eta \rho |I|$ samples with $\kappa = 1 + O(\delta)$; this set has no Nyquist gaps, so by Beurling $\sum_{\text{shallow} \cup \text{on}} |F(t_\rho)|^2 \geq c \eta \rho \|F\|_2^2$. Taking δ small gives $\mathbf{t}_+(F) \geq c' \rho \|F\|_2^2$.*

4.1. The honest status of (H). (H) is strictly weaker than RH: RH says there are *no* off-line zeros; (H) says only that deep ones do not *locally dominate* (they may exist, even infinitely many, provided they do not fill a Nyquist cell). It is implied by RH and by any uniform short-interval bound $N(\frac{1}{2} + \delta, T + \frac{1}{d}) - N(\frac{1}{2} + \delta, T) = o(\rho)$.

However, (H) is *not* established unconditionally by current methods. The global density $N(\frac{1}{2} + \delta, T) \ll T^{1-c\delta}$ leaves a budget allowing $\sim \log T$ deep zeros to be concentrated in a single short interval (cost $\log T \ll T^{1-c\delta}$), which the global bound does not forbid. Selberg’s fluctuation theory makes such a cell-filling cluster a large deviation of S , of measure $\ll \exp(-c(\log T)^2 / \log \log T)$ —extraordinarily rare, but not provably absent at sporadic large heights. And (H) is necessary: without it there exist F with $\mathbf{t}_+(F) = 0$ but $\mathbf{t}_-(F) \neq 0$, breaking closability in H_+ . Alternative realizations (all-zeros norm, L^2) reduce to the same gap/clustering condition. Hence **Theorem 1.1 is unconditional modulo (H), with (H) a uniform short-interval zero-density bound far weaker than RH, very likely true, but at the frontier of the unconditional.**

5. THEOREM A: THE SEMIBOUNDED REALIZATION

Proof of Theorem 1.1. By Lemmas 3.4–3.5 and Lemma 4.2 (under (H)), $\tilde{\mathfrak{t}}_- \leq \frac{C_d}{c} \mathfrak{t}_+ =: C \mathfrak{t}_+$, hence $|\mathfrak{t}_-| \leq 4C \mathfrak{t}_+$ (the factor 4 is removed in §7). In H_+ , where $\mathfrak{t}_+ = \|\cdot\|_{H_+}^2$,

$$\frac{\mathfrak{t}(F)}{\|F\|_{H_+}^2} = 1 + \frac{\mathfrak{t}_-(F)}{\mathfrak{t}_+(F)} \geq 1 - 4C.$$

$\mathfrak{t}_+$ is closed (a monotone limit of bounded forms; Kato [6]), and $\mathfrak{t}_-$ is form-bounded relative to it with bound $4C$, so $\mathfrak{t} = \mathfrak{t}_+ + \mathfrak{t}_-$ is closed and semibounded; $\mathcal{T}$ is the self-adjoint operator associated to it via KLMN, with form domain H_+ . $\square$

Corollary 5.1 (Faithful reformulation). *The positivity $\mathfrak{t} \succeq 0$ does not refer to the inner product, so its truth is norm-independent; with Theorem 1.1,*

$$\boxed{\text{RH} \iff \inf \text{spec}(\mathcal{T}) \geq 0}$$

—*RH is exactly the sign of the finite bottom of the rigorously realized operator $\mathcal{T}$.*

The magnitude estimates give $C \geq e^d > 1$, so the bottom $1 - 4C$ is negative; lowering C to $\leq \frac{1}{4}$ is exactly RH. Theorem 1.1 relocates RH to the sign of a concrete finite object; it does not evaluate that sign.

6. THEOREM B (1): THE KREĬN IDENTITY

This section is unconditional. Define the evaluation operator $E : PW_d \rightarrow \ell^2(\rho)$, $(EF)_\rho := F(t_\rho + ib_\rho)$ (bounded by Lemma 3.5), and the fundamental symmetry J as the linear involution induced by the partner map (3), $(Jc)_\rho := c_{\sigma(\rho)}$. Then $J^2 = I$ and $J = J^*$ (a permutation by an involution is self-adjoint): a genuine Kreĭn fundamental symmetry.

Proposition 6.1 (The identity). *With $\langle a, b \rangle = \sum_\rho a_\rho \overline{b_\rho}$,*

$$\langle EF, JEF \rangle = \sum_\rho (EF)_\rho \overline{(EF)_{\sigma(\rho)}} = \sum_\rho F(t_\rho + ib_\rho) \overline{F(t_\rho - ib_\rho)} = \mathfrak{t}(F),$$

i.e. $\boxed{\mathfrak{t} = E^* J E, \quad J = J^*, \quad J^2 = I} \quad (\text{exact, unconditional}).$

Proof. Immediate from $b_{\sigma(\rho)} = -b_\rho$, $t_{\sigma(\rho)} = t_\rho$ and (4). $\square$

Remark 6.2. Writing J with a complex conjugation, $(Jc)_\rho = \overline{c_{\bar{\rho}}}$, makes it antilinear; that coincides with the linear J above on real-type test functions $F(\bar{z}) = \overline{F(z)}$. The linear involution is the convention-free statement.

Three structural consequences.

- (C1) $RH \Rightarrow J = I \Rightarrow \mathcal{T} = E^* E \geq 0$. Under RH all $b_\rho = 0$, so $\sigma(\rho) = \rho$, $J = I$, $\mathfrak{t} = E^* E \geq 0$: Weil positivity becomes the trivial positivity of a Gram operator.
- (C2) $RH \text{ fails} \Rightarrow J = I_{\text{on}} \oplus \bigoplus_{\text{pairs}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Each off-line quartet splits under σ into two swap-pairs $\{\rho, 1 - \bar{\rho}\}$, each a 2×2 swap with spectrum $\{+1, -1\}$. All negative directions of J come exactly from off-line zeros.
- (C3) $RH \iff E^* J E \geq 0 \iff R(E) \text{ is a } J\text{-non-negative subspace.}$

The conceptual gain is the clean separation of the geometry of PW_d (through $R(E)$), the location of zeros (through σ , hence J), and the sign (through the indefinite form $[\cdot, \cdot]_J$).

7. THEOREM B (II): THE ANGULAR OPERATOR AND THE SINGLE AXIS

Let $\ell^2(\rho) = \mathcal{K}_+ \oplus \mathcal{K}_-$ with $J = P_+ - P_-$. On a swap-pair the ± 1 eigenvectors are $\frac{1}{\sqrt{2}}(e_\rho \pm e_{\sigma\rho})$, so with $F_\pm := F(t_\rho \pm ib_\rho)$,

$$\|P_+EF\|^2 = \mathfrak{t}_+ + \sum_{\text{pairs}} \frac{1}{2}|F_+ + F_-|^2, \quad \|P_-EF\|^2 = \sum_{\text{pairs}} \frac{1}{2}|F_+ - F_-|^2,$$

and exactly (no factor-4 slack)

$$(6) \quad \boxed{\mathfrak{t}(F) = \|P_+EF\|^2 - \|P_-EF\|^2.}$$

Lemma 7.1 (Qualitative injectivity; unconditional, not circular). $P_+EF = 0 \Rightarrow F = 0$.

Proof. $P_+EF = 0$ forces F to vanish at all on-line zeros. By the type-density uniqueness theorem (a nonzero L^2 function of exponential type d cannot vanish on a real sequence of density $> d/\pi$ [8]), and the on-line zeros have density $\frac{1}{2\pi} \log T \rightarrow \infty \gg d/\pi$, we get $F = 0$. This uses only the zeros of F , no sampling/coercivity input. $\square$

Hence $\overline{R(E)}$ is the graph of the *angular operator* $K := P_-(P_+|_{R(E)})^{-1} : \mathcal{K}_+ \supseteq \text{dom } K \rightarrow \mathcal{K}_-$, defined unconditionally (possibly unbounded). Krein theory [1, 2] gives the single-axis dictionary

$$(7) \quad \underbrace{\text{type-density}}_{K \text{ defined}}; \quad \underbrace{(H) \Rightarrow \|K\| < \infty}_{\text{B-2 (finite bottom)}}; \quad \underbrace{\|K\| \leq 1 \iff \text{RH}}_{\text{given } K \text{ defined}}.$$

Here $\|K\| < \infty \iff R(E)$ uniformly positive-projecting $\iff$ the coercivity B-2 needs; (H) is sufficient for it (the honest direction is $(H) \Rightarrow \|K\| < \infty \Rightarrow \text{B-2}$). And $\|K\| \leq 1 \iff \|P_-EF\| \leq \|P_+EF\| \forall F \iff \mathfrak{t} \geq 0 \iff \text{RH}$. From Lemma 3.4, $\|K\| \geq e^{d/2} > 1$ on the magnitude side—the wrong side of 1.

Cleanest statement of this line of work. B-2 is “ K bounded” (needs (H)); RH is “ K a contraction.”

7.1. Where the negativity lives, and a warning. On a swap-pair the negative coordinate is the vertical antisymmetric difference

$$d_\rho := F(t_\rho + ib_\rho) - F(t_\rho - ib_\rho) = 2ib_\rho F'(t_\rho) + O(b_\rho^3 \|F''\|),$$

so to leading order $\|P_-EF\|^2 \approx \sum_{\text{pairs}} 2b_\rho^2 |F'(t_\rho)|^2$: the negative directions are depth-weighted derivative samples, and $\|K\| \rightarrow 0$ as depths $\rightarrow 0$.

Remark 7.2 (The optimistic-pillar trap). The b_ρ^2 form is the *shallow linearization only*. The Taylor remainder is $O(b_\rho^3 \|F''\|)$ with $\|F''\| \leq d^2 \|F\|$ (Bernstein); for *deep* zeros ($|b_\rho|$ up to $\frac{1}{2}$, d moderately $> \frac{1}{2}$) the remainder/leading ratio is $\sim b_\rho^2 d = O(1)$ —the expansion does not converge, and the true vertical evaluation carries the $e^{2d|b_\rho|}$ amplification responsible for $\|K\| \geq e^{d/2}$. Thus a “ $\sum b_\rho^2 |F'|^2 \leq \theta \mathfrak{t}_+ \Rightarrow \text{RH}$ ” inequality would deliver RH only on the shallow part already known harmless. The correct concrete target keeps the full vertical difference:

$$(8) \quad \|K\|^2 = \sup_F \frac{\sum_{\text{pairs}} \frac{1}{2} |F(t_\rho + ib_\rho) - F(t_\rho - ib_\rho)|^2}{\mathfrak{t}_+ + \sum_{\text{pairs}} \frac{1}{2} |F(t_\rho + ib_\rho) + F(t_\rho - ib_\rho)|^2}, \quad \|K\| \leq 1 \iff \text{RH}.$$

8. THEOREM C: A NO-GO THEOREM FOR THE ZERO-SIDE ATTACK

Proof of Theorem 1.3. The pivot is that the explicit formula gives $\mathfrak{t}(F)$ as the *same* functional in two ways:

$$(9) \quad \underbrace{\|P_+EF\|^2 - \|P_-EF\|^2}_{\text{zero side: depends on } \{t_\rho, b_\rho\}} = \mathfrak{t}(F) = \underbrace{\mathfrak{a}(F) + \mathfrak{q}(F) - \mathfrak{p}(F)}_{\text{arithmetic side: contains no } b_\rho}.$$

(i) **No independent information in the depths.** The arithmetic side sums the depths out—it does not contain b_ρ . Since both sides equal the same functional of F , any bound on $\|K\|$ phrased through the b_ρ is, by (9), a bound on $\mathfrak{t}$ through the arithmetic side in disguise.

(ii) **Tautology w.r.t. Weil.** $\|K\| \leq 1 \iff \|P_-EF\| \leq \|P_+EF\| \forall F \iff \mathfrak{t} \geq 0 \iff \mathfrak{a} + \mathfrak{q} - \mathfrak{p} \geq 0 \iff \text{RH}$. Lowering $\|K\|$ to ≤ 1 is proving arithmetic positivity.

(iii) **Sign-agnostic bounds land on the wrong side.** Dropping the phase cancellation between F_+ and F_- (triangle inequality $|F_+ - F_-|^2 \leq 2(|F_+|^2 + |F_-|^2)$) gives $\|P_-EF\|^2 \leq \kappa \cdot (\text{off-line cell count})\|F\|^2$, κ up to $e^{2d|b|}$, hence $\|K\|^2 \gtrsim e^d$. Retaining the phase requires the exact analytic continuation of F at $t_\rho \pm ib_\rho$; for deep zeros that is not the $2ib_\rho F'$ linearization but the global behaviour of F —the arithmetic side again. $\square$

Corollary 8.1 (Why the zero-side realization is circular as a tool). *To even write $K = P_-(P_+|_{R(E)})^{-1}$ one must know the off-line zero data $\{(t_\rho, b_\rho) : b_\rho \neq 0\}$ —the very data whose emptiness is RH. The realization is a true identity but presupposes the unknowns. The Connes–Consani realization is the useful one precisely because its operator is built from explicit archimedean data, independent of the zeros. This is the structural reason the zero-side picture cannot attack the sign.*

No-go: $\|K\|$ depends on the off-line depths only through $\mathfrak{t}$;
bounding $\|K\| \leq 1$ from the zero side $\equiv$ Weil positivity $\equiv$ RH.

9. RELATION TO PRIOR WORK

Connes–Consani [4]. They root Weil positivity in the *compression of the scaling action* ϑ on $L^2(\mathbb{R})_{\text{ev}}$ onto Sonin’s space $\mathbf{S}$ (the orthogonal complement of the range of the phase-space cutoff projection), proving $W_\infty(g * g^*) \geq \text{tr}(\vartheta(g) \mathbf{S} \vartheta(g)^*)$ and controlling the residual difference (between the Weil distribution and the Sonin trace) via prolate spheroidal wave functions and Hermitian Toeplitz matrices, by an eigenvalue-below-1 argument. Our reduction “ $\|K\| \leq 1 \iff \text{RH}$ ” is of the same type (a compression with a norm/eigenvalue-below-1 positivity criterion). We do *not* claim an identification $U K U^{-1} = K_{\text{CC}}$: the two are distinct realizations of the same form $\mathfrak{t}$ —ours *zero-side* (depends on the zeros), theirs *archimedean* (explicit, independent of the zeros). Theorem 1.3 and Corollary 8.1 explain the asymmetry: the archimedean realization is a genuine handle, the zero-side one is circular.

Suzuki [11]. Completing the Weil hermitian form yields, *under RH*, a de Branges space $\mathcal{H}(E_\xi)$ with $E_\xi(z) = \xi(\frac{1}{2} - iz) + \xi'(\frac{1}{2} - iz)$ (Hermite–Biehler under RH [7]), turning Weil’s inequality into equalities. Suzuki notes (after Selberg) that “the construction of a space assuming RH will not be useful for attacking RH” and therefore builds unconditional spaces $\mathcal{H}_0, \mathcal{K}_0$ via screw lines. Our completion is the same in spirit but *directly* unconditional: with off-line zeros allowed, the metric is genuinely indefinite, so the natural object is a *Kreĭn* space (Pontryagin only if the off-line set were finite, which is not known). The Kreĭn-vs-de Branges dichotomy is precisely the positivity-built-in vs not distinction; our Theorem 1.3 is the rigorous form of Selberg’s warning, specialized to the zero-side realization.

Classical lineage. Weil’s criterion [12, 3]; de Branges’ *Hilbert Spaces of Entire Functions* [5] (the positive regime); Azizov–Iokhvidov [1] and Bognár [2] for Kreĭn-space angular operators; Montgomery [9] and Selberg [10] for the zero statistics behind (H). Our place is a modular reformulation plus a no-go, not a new mechanism for positivity.

10. HONEST STATUS AND OPEN PROBLEMS

Statement	Status
Lemmas 3.1–3.5 (IBP, quadrature $O(t_+)$, vertical, P–P)	unconditional
Depth split; (H) $\Rightarrow$ coercivity (Lemma 4.2)	unconditional
Theorem 1.1 (semibounded $\mathcal{T}$, $\inf \text{spec} \geq 1 - 4C$)	modulo (H)
Corollary 5.1 (RH $\iff$ sign of the bottom)	modulo (H)
Theorem 1.2 (i): $t = E^*JE$	unconditional
Theorem 1.2 (ii): angular operator, dictionary (7)	unconditional
Theorem 1.3: no-go for the zero-side attack	unconditional
RH itself (the sign)	not proved; Thm 1.3 shows this route cannot

What is not claimed. No proof of RH. Theorem 1.1 is conditional on (H); (H) is much weaker than RH but not unconditionally established (§4.1). The Kreĭn picture is the correct language and the precise target, and Theorem 1.3 proves it is not a route past the sign.

Open problems.

1. *Is (H) unconditional?* Is the uniform (every cell, all large T) short-interval zero-density for fixed $\sigma > \frac{1}{2}$ unconditional, or only on average? This is the one item separating “unconditional B-2” from “B-2 modulo (H)”.
2. *Sharp constants.* The Plancherel–Pólya constant of Lemma 3.5 with clustering and margin $d - \frac{1}{2}$; the quantitative Beurling step in Lemma 4.2.
3. *The only route to the sign.* By Theorem 1.3 the sign cannot come from the zero side; it requires *structural positivity*—a factorization $\mathcal{T} = A^*A$, Connes’ archimedean trace positivity, or a de Branges chain—an independently-given positive structure, not a bound. This is the genuine RH problem and is left open. It is the subject of the companion research plan (route B).

Conclusion. Starting from Weil’s criterion (RH $\iff$ a positivity), we built a faithful, RH-independent, semibounded operator realization $\mathcal{T}$ whose bottom is finite and whose sign is RH (Theorem 1.1), exposed its exact Kreĭn structure $t = E^*JE$ and the single controlling axis $\|K\|$ (Theorem 1.2), and proved that this zero-side structure—however explicit its dependence on the off-line zeros—cannot decide the sign, because the depths enter only through the very functional whose positivity is RH (Theorem 1.3). The program therefore delivers a precise localization of RH’s difficulty (the contraction $\|K\| \leq 1$) and a rigorous account of why the natural zero-side attacks fail—but not a proof. The sign, as long suspected, requires structural positivity, which the magnitude/zero-side machinery provably cannot supply. This closes the explicit-formula/zero-side line of this line of work.

REFERENCES

- [1] T. Ya. Azizov, I. S. Iokhvidov, *Linear Operators in Spaces with an Indefinite Metric*, Wiley, Chichester, 1989.
- [2] J. Bognár, *Indefinite Inner Product Spaces*, Springer, 1974.
- [3] E. Bombieri, *Problems of the Millennium: The Riemann Hypothesis*, Clay Mathematics Institute, 2000.
- [4] A. Connes, C. Consani, *Weil positivity and trace formula, the archimedean place*, Selecta Math. (2021); arXiv:2006.13771.
- [5] L. de Branges, *Hilbert Spaces of Entire Functions*, Prentice-Hall, Englewood Cliffs, 1968.
- [6] T. Kato, *Perturbation Theory for Linear Operators*, 2nd ed., Springer, 1976.
- [7] J. C. Lagarias, *Hilbert spaces of entire functions and Dirichlet L-functions*, in *Frontiers in Number Theory, Physics, and Geometry I*, Springer, 2006, 365–377.
- [8] B. Ya. Levin, *Lectures on Entire Functions*, Transl. Math. Monographs 150, Amer. Math. Soc., 1996.
- [9] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, in *Analytic Number Theory*, Proc. Sympos. Pure Math. 24 (1973), 181–193.
- [10] A. Selberg, *On the remainder in the formula for $N(T)$, the number of zeros of $\zeta(s)$ in the strip $0 < t < T$* , Avh. Norske Vid. Akad. Oslo (1944).
- [11] M. Suzuki, *On the Hilbert space derived from the Weil distribution*, arXiv:2301.00421 (v3, 2025).

- [12] A. Weil, *Sur les “formules explicites” de la théorie des nombres premiers*, Comm. Sém. Math. Univ. Lund (1952), 252–265.